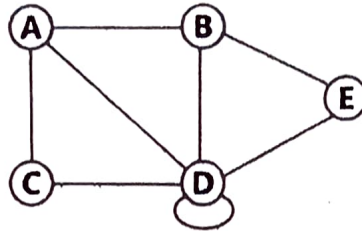


Bachelor Of Engineering In Information Technology
2nd Year 2nd Semester, Semester Examination, 2023
Subject Name –Graph Theory & Combinatorics (IT/PC/B/T/224)

Full Marks=100

CO1
[10]

Q1. (i) Find the adjacency matrix of the graph given below. Find out the number of paths of length 2 from one vertex to others.



(ii) Find out the maximum and minimum number of edges in a 3-component graph having 10 vertices.
 [5+5=10]

CO2
[20]

Q2. (i) Does a Hamiltonian path or circuit exist on the graph below (Figure: 1)? Justify it.
 (ii) Prove with a suitable example "Every tree has either one or two centers".
 (iii) Using Prim's algorithm find the minimal spanning tree of the graph given below (Figure: 2).

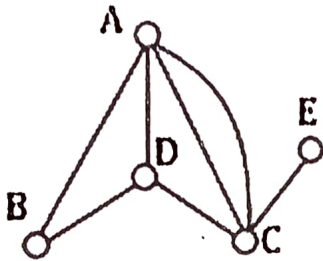


Figure: 1

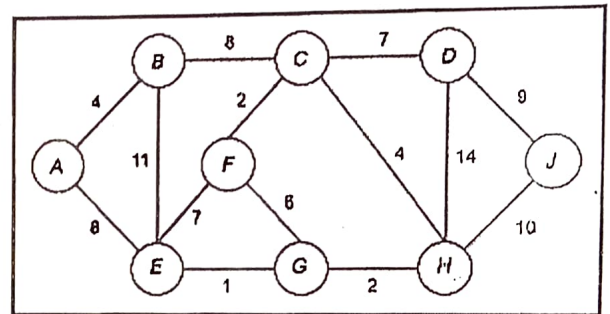


Figure: 2

[6+6+8=20]

CO3
[20]

Q3. (i) Does a cut point and cut edges exist of the graph given below (Figure: 3)? If yes then mention them.
 (ii) Find the union of the two graphs G_1 and G_2 given below (Figure: 4).
 (iii) Justify this using a suitable example "The vertex connectivity of any graph G can never exceed the edge connectivity of G ".

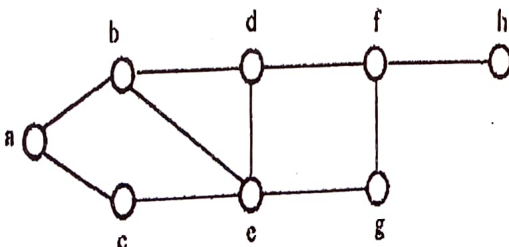


Figure: 3

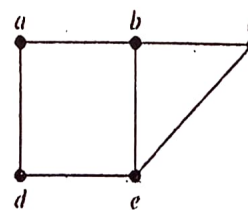
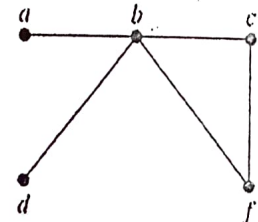
 G_1  G_2

Figure: 4

[7+7+6=20]

CO4
[20]

Q4. (i) "A graph is 2-colorable if it is bipartite and every cycle has an even length." Justify this with a suitable example.
 (ii) If a graph has 100 vertices and 300 edges, can it be planar?
 (iii) Find the chromatic number of the graph given below (Figure: 5). (Mention all steps properly)

(iv) Find out how many vertices can be matched using maximum matching in the bipartite graph algorithm of the following graph given below (Figure: 6)? Can we call it perfect matching? Justify this.

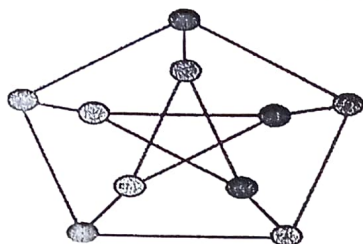


Figure: 5

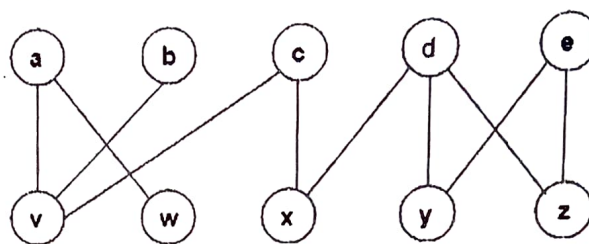


Figure: 6

[6+3+4+(4+3)=20]

CO5
[20]

Q5. (i) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three hearts are selected??

(ii) How many ways are there for **eight men** and **five women** to stand in a line so that **no two women** stand next to each other?

(iii) Suppose a club has 25 members. How many ways are there to choose a **president**, **vice-president**, **secretary**, and **treasurer** of the club, where no person can hold more than one office?

[6+7+7=20]

CO6
[10]

Q6.(i) Find the generating functions of the following sequences given below in closed form .

$$0, 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots$$

(ii) Using generating functions, find a_n in terms of n for the case given below:

$$a_0 = 1, a_1 = 2 \text{ and } a_{n+2} = 5a_{n+1} - 4a_n \text{ for } n \geq 0$$

[5+5=10]

CO1: Explain and discuss the concept of different types of Graphs with fundamental properties and express different types of matrix representation. (K2)

CO2: Illustrate different types of trees such as (i) rooted tree (ii) spanning tree etc, and explain their properties. (K3)

CO3: Apply operations like Union, Deletion, and decomposition of graphs and illustrate Cut vertex and Cut edge and their properties. (K3)

CO4: Illustrate planar graph and their properties and Graph Coloring and Matching. (K3)

CO5: Apply and evaluate basic counting rules, pigeon-hole principle and principle of inclusion-exclusion. (K3)

CO6: Apply and Solve problems using Generating Function and Recurrence Relations. (K3)